

Machine Learning–Based Dengue Diagnosis Using Clinical and Routine Hematological Data Without Antigen Testing: Evidence from Bangladesh

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Abstract. Dengue fever is a significant public health issue for regions with limited diagnostic procedures. Although the conventional lab-based methods are reliable, they are usually costly and time-consuming. In the research, routine clinical features and simple hematological parameters (platelet and white blood cell counts) are used to predict dengue fever without using antigen tests or serological testing on the patient. The method offers an easy, cheaper and more expandable system for early dengue detection, based on the common symptoms and basic clinical signs. This paper compares various Machine Learning (ML) classifiers, which are Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), K-Nearest Neighbors (KNN), Gradient Boosting (GBM) and Support Vector Machine (SVM) on structured data of 1,018 patient records, which was collected from a hospital in Bangladesh. An optimized Random Forest model, which includes data balancing and parameter tuning was evaluated alongside base classifiers. To measure robustness, another dataset outside the primary dataset was harmonized with the primary dataset and the same model was re-evaluated with this combined dataset. The optimized Random Forest had 99.02% as the accuracy (AUC 0.99) for the primary data and 90.49% (AUC 0.994) accuracy for the merged dataset indicating high predictive performance and generalizability with variably different data. This study proves that an ML-based diagnostic system can enable dependable, inexpensive, and rapid decision support in the early detection of dengue without antigen tests.

Keywords: Dengue Fever · Machine Learning · Early Diagnosis · Random Forest · SMOTE · Clinical Data.

1 Introduction

Dengue fever is an infectious disease spread by mosquitoes and caused by four types of viruses (DENV-1 to DENV-4). Humans are mainly affected by its

transmission during bites of different *Aedes* species. According to the World Health Organization (WHO), 390 million cases of dengue infection are experienced worldwide annually, out of which about 96 million are clinically manifested [1]. Symptoms may be mild such as high fever, rash, headache, and pain in muscles or may be severe and likely to be life-threatening, like DHF and DSS [2]. Early diagnosis of dengue is essential for controlling the disease and reducing deaths, but it is difficult due to the similarity with other fevers like malaria, chikungunya, and influenza [3]. Conventional procedures like ELISA and PCR, although costly and time-consuming, remain the industry standard. This has led to the attraction of machine learning that provides faster, cheaper and scalable predictions based on borrowed data [4].

Machine learning would be an effective tool to predict dengue fever and severity since it can model complicated and non-linear relationships in the large dataset. In recent studies, it has been established that models like the SVM, DT, LR, RF and deep learning frameworks can be effective in the diagnosis of dengue and prediction of outbreaks [5].

This study offers the comparison of various multiple ML classifiers (LR, SVM, DT, RF, KNN and GBM) on a structured dataset containing confirmed dengue and non-dengue cases. Once the baseline is achieved, a custom Random Forest model is built, with class balancing using SMOTE, standardized preprocessing, and hyperparameter tuning by using Grid Search with stratified 5-fold cross-validation (CV). The stability of the model is also tested with an external dataset that is harmonized and merged with the primary dataset to allow consistencies and then taken through the same preprocessing and evaluation pipeline.

We aim for detection in the early clinical stage based on common symptoms alone and minimal blood testing. These diagnoses are given labels based on clinician’s documented decisions rather than on confirmations in a laboratory. We construct our model for assistance in clinical triage in resource-poor settings. We establish high sensitivity so suspected cases would actually receive confirmatory testing as well as increased follow-up.

2 Related Work

Abdualgalil et al. [4] suggested an ML solution for the early dengue detection with the help of clinical data (2017–2019) of the Epidemiological Monitoring Centre in Taiz, Yemen. They compared KNN, GBC, ETC, XGB and LightGBM, and ETC performed the best: 99.12 for the criterion of holdout and 99.03 for cross-validation of 10 folds, evaluated in F1-score, accuracy, precision, AUC, recall and operation time.

Gautam et al. [5] proposed Anti-Dengue, a QSAR-based ML model to predict small molecule inhibitors of the dengue virus in support of drug repurposing. The modelling was carried out using 238 compounds of DrugRepV and a total of more than 2000 drugs in DrugBank with four models (SVM, ANN, KNN and RF) being tested and SVM performing best (train/test PCC 0.71, validation PCC 0.81).

Chaw et al. [6] trained the ML models to predict shock occurrences in patients with dengue based on the data in the ICUs of 170 hospitals in Malaysia. In logistic regression, decision tree, ANN, SVM, and ensembles, bagging was found to provide the best results (AUC 0.79), and hemoglobin (Day 2), INR, and age were shown as critical predictors.

Hussain et al. [7] have reviewed the dengue forecasting algorithms such as decision tree, SVM, neural network and evolutionary algorithms. They observed that adding clinical, demographical and environmental information to prediction increases accurate prediction. The other important feature of the study was the cooperation of health practitioners and data scientists in disease management.

Iqbal et al. [8] performed a comparison of seven machine learning models for predicting dengue outbreaks, utilizing 75 cases with eight clinical characteristics. LogitBoost performed best—92% accuracy, 90% sensitivity, and 94% specificity—and surpassed the accuracy and least percent error of logistic regression, decision trees, and others.

AutoTiC-NN, as suggested by Appice et al. [9], is a multistage ML model that predicts the cases of dengue disease regarding temperature and the previous incidents of dengue in around 32 states of Mexico throughout the years 1985–2010. The clustering approach and dimension reduction through autoencoding, as well as nearest-neighbor prediction, are used in the methodology.

Davi et al. [10] predicted dengue severity using genetic data on 102 genetic Brazilian samples containing 322 SNPs. They used SVM as a feature selection and ANN as a classifier. The model had a sensitivity of 98%, an accuracy of 86%, and a specificity of 51%, indicating that genetics are relevant in the severity of dengue.

Zhao et al. [11] forecasted dengue cases in Colombia based on Random Forest and ANN. Random Forest was more accurate and had mean absolute errors of 9.32 (1 week) to 24.56 (12 weeks). It presented smaller relative errors as compared to ANN, providing better forecasts.

Sanjudevi et al. [12] applied SVM and Decision Tree to forecast dengue using a UCI dataset (108 cases, 18 features). SVM achieved 99% accuracy after the feature reduction using stepwise regression compared to the Decision Tree (87.5%). Sensitivity, specificity and AUC scores were used to evaluate performance.

3 Methodology

3.1 Overview

This study develops a customized machine learning pipeline for analyzing the classification of dengue fever cases using clinical data. Fig. 1 shows the methodological workflow. To start with, several pre-trained models (DT, SVM, LR, KNN) and a baseline random forest were trained on the formatted data and the results of their classification accuracy were recorded as a baseline measure of performance.

A custom Random Forest Classifier was created with the feature selection, SMOTE-based balancing, preprocessing, and hyperparameter tuning through

5-fold grid search cross-validation. To identify performance improvement, its accuracy was compared with pre-trained models. A different dataset was then harmonized, encoded and cross-validated to ensure the model was consistent, robust and generalizable.

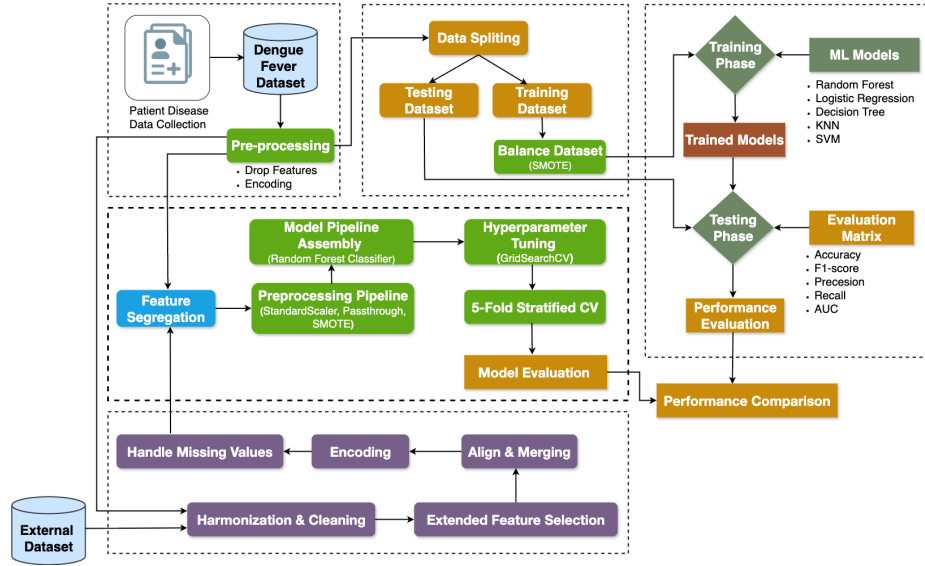


Fig. 1. Proposed Methodology Diagram

3.2 Dataset Description

The research is based on a clinical dataset in Life Aid General Hospital and Diagnostic LTD, Munshiganj, Dhaka, that has 1,018 patient records of suspected dengue cases. These were 697 dengue-positive and 321 negative based on clinical and symptom assessment. No patient identifiable information was captured and this guarantees privacy and ethical adherence. The dataset contains no missing values; therefore, the dataset is appropriate to be used in statistical analysis and machine learning. All of the variables contained in the dataset are listed and explained in Table 1.

3.3 Data Pre processing

Columns with no informative data were discarded such as ID and Location columns for reducing noise and maximizing the performance of the model. Such attributes were deemed unnecessary in the classification process since they made no contribution to the prediction of the model's power. Label encoding was

Table 1. Dataset Overview.

Serial No.	Feature Name	Type
1	Gender	categorical
2	Age	numerical
3	Platelet Count	numerical
4	Fever	categorical
5	Location	categorical
6	WBC	numerical
7	Duration of Fever	categorical
8	Headache	categorical
9	Muscle Pain	categorical
10	Vomiting	categorical
11	Rash	categorical
12	Outcome	categorical

performed to transform categorical variables into a numerical format and could be used as input for the machine learning method. The categorical variables like Gender, Fever, Headache, Rash, Vomiting, Muscle Pain, and Outcome are encoded with unique integer values. Encoding was done by assigning a mapping to any distinct category v_i by a particular integer i such that the equation is:

$$f(v_i) = i \quad \text{for } i \in \{0, 1, \dots, n-1\} \quad (1)$$

3.4 Data Splitting

Data was split 80:20 for training and testing set. The training set was used to provide information of the input features and the output labels that was utilized to train the model, whereas the test set was employed to make a fair judgement of the generalizability of the model.

3.5 Training set balancing

The training set was imbalanced (557 dengue-positive vs. 257 dengue-negative), this might favor the majority group in the model. We used SMOTE, a technique that generates additional data points for the minority class by combining existing samples with their nearest neighbors. Timely diagnosis of dengue is essential for controlling the disease and reducing. Sampling strategy we used was minority, k neighbors = 5 and random state = 42. The training set was resampled and balanced at 557 vs. 557, meaning that 300 synthetic samples were sampled in the minority class (see Fig. 2). Notably, resampling was only used on the training data; all test sets were used with their original class distributions.

3.6 Model Training and Evaluation Pipeline

This section describes the training and testing procedure of various ML models such as LR, DT, RF, SVM and KNN on the training set. All models will be



Fig. 2. Dataset distribution before and after applying SMOTE

evaluated according to precision, accuracy, F1-score, recall, and AUC using a testing set.

Training Phase. The training phase refers to the initial training of models through machine learning (ML) and the training dataset. This stage is very important because this is where the model learns the patterns and relationships amongst the features (X) to its associated target labels (y) in the training set.

Machine Learning Models. Various ML models were used to produce the probability of the dengue fever which include the DT, RF, LR, SVM and KNN. These models were chosen because they can classify different types of data. The characteristics of each of the models employed are different and the justification behind their choice and the use of the model is explained as follows.

Testing Phase. During the testing phase, the trained models are tested through a test using the testing dataset. During training, the testing dataset is not used. It is a separate collection of data that the models have never seen before. This step is very important for figuring out how well the model works with data it hasn't seen before.

3.7 Custom Random Forest Model Development

The trained Random Forest Classifier was customized according to their response to the pre-trained models to predict dengue disease. Feature segregation, splitting the data and standard scaler preprocessing are present in the model. It uses SMOTE to resample the training data and Grid Search with cross-validation to optimize hyperparameters. Our performance is evaluated by the 5-fold cross-validation and is rated using such measures as precision, accuracy, F1 score, recall, and AUC.

Feature Segregation. Feature segregation means distinguishing input variables (X) and the target variable (y). The independent variables represent the features, and what should be predicted is the target. For example, in the prediction of dengue, the feature might be clinical and demographic data, and the target is whether or not the patient has dengue.

Pre-processing Pipeline. The preprocessing pipeline is highly significant for preparing the data to train a model. It normally includes such processes as scaling the features such that they are placed on the same scale which makes the model learn.

Standard Scaler: It is just the standard type of scaling where the feature dimensions are transformed to that of a mean zero and a variance one invariant scale. This is significant to enable algorithms such as Random Forest to work best although the algorithms do not tend to be as sensitive in regard to feature scaling as much as other algorithms.

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma} \quad (2)$$

Where X denote the feature matrix, μ denote the mean of each feature and σ is the standard deviation.

Passthrough: This step passes any additional data through without changing it. If there are any features that do not require transformation, they are left unchanged.

SMOTE: Once again, data SMOTE was used to balance the data. In section 3.5 a detailed explanation of the dataset balancing technique is described.

Model Pipeline Assembly. The model pipeline assembly is the process of placing together a pipeline that includes all the stages needed for preprocessing and training a model. The model work-flow for this custom Random Forest classifier is as follows:

Random Forest Classifier: Random Forest is a way of learning that makes a group of decision trees and combines their predictions. Random subsets of the data and features are used to build each tree. The model combines the outputs of all the trees to make the final guess.

Hyperparameter Tuning. Tuning was conducted with the use of Grid Search and Stratified 5-Fold CV in order to determine the best configuration for Random Forest Classifier.

5-Fold Stratified Cross-Validation. We performed a stratified 5-fold CV on primary data. Preprocessing was fit on each training split in a fold, then SMOTE was applied on such training split; the validation holdout split remained untouched (natural class distribution). Such a step guarantees no information

leakage for any synthetic examples in any validation or test set. The overall accuracy was obtained by summation of all the folds as:

$$\text{Accuracy}_{\text{CV}} = \frac{1}{5} \sum_{i=1}^5 \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i} \quad (3)$$

The approach reduced bias and permitted the performance to be estimated with stability and was capable of comparing with Grid Search to fine tune hyperparameters.

Model Evaluation. Once the Random Forest Classifier was trained with the best hyperparameters, its performance was assessed with a few metrics to gain insight into it. It has been tested on the testing set and the core goal was to determine how well the model could predict both classes (dengue-positive and dengue-negative).

Accuracy: Accuracy was obtained as the ratio of the number of correct predictions out of the total correct messages:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

Where TP , TN , FP and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

Precision: Precision can be thought of as the accuracy of the number of actual positive predicted against the number of total possible positives predicted:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

Recall: It is the proportion of the positive prediction which is correctly predicted to all the positive actual predictions:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

F1-score: To gain a balanced measure of performance of the model F1-Score, the harmonic mean of recall and precision was calculated:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

AUC (Area Under the ROC Curve): AUC was computed to compare the capacity of the model to differentiate between those two classes in various thresholds. The AUC close to 1 means a high capacity to distinguish between the disease dengue-positive and dengue-negative.

3.8 External Dataset Integration and Pre-processing pipeline for Model Validation

To check the robustness and generalization power of the developed Random Forest Classifier, the external dataset was merged to the primary dataset used as the dengue fever dataset. It has inconsistent column labels, the availability of features, and the form of values. This required numerous preprocessing stages to harmonize, clean, and align the two datasets in order to then merge the data. Then the dataset was merged for model training and evaluation. The pre-processing process consisted of the harmonization and cleaning, the long feature selection, schema alignment and merging, labeling and missing values.

Harmonization and Cleaning. Harmonization was done to align naming schemes in datasets. The column names in the external dataset were corrected to be consistent; for instance:

Sex → Gender, Total WBC count(/cumm) → WBC.

Sex → Gender, Result → Outcome.

Outcome labels in both datasets were standardized by capitalizing the first letter:

“positive” → “Positive”, “negative” → “Negative”.

Similarly, the Gender field was transformed to lowercase in both datasets to provide consistency before encoding:

“Male” → “male”, “Female” → “female”.

This method helps avoid inconsistencies in categorical values when performing the merging process.

Extended Feature Selection. After harmonization, only the relevant features were retained: the external dataset contributed blood test parameters, while the primary dataset added symptoms and clinical data, ensuring consistency and keeping variables essential for dengue diagnosis.

Alignment and Merging. Since the chosen sets of features were not equal in the datasets, the missing columns were added to each of the datasets with null values to have the same schema before the merge. Mathematically, if:

$$C_1 = \text{columns in original dataset}, \quad C_2 = \text{columns in external dataset} \quad (8)$$

Then for all $c \in C_1 - C_2$, a null column was added to the external dataset, and for all $c \in C_2 - C_1$, a null column was added to the original dataset. Then, the two datasets were merged row-wise:

$$D_{\text{merged}} = D_{\text{original dataset}} \cup D_{\text{external dataset}} \quad (9)$$

Encoding. Label encoding was done once more on the merged dataset. The detailed description of label encoding is provided in the 3.3 section.

Missing Value Handling. Missing values are replaced according to the type of variables and this avoids bias and also does not tamper with the integrity of the dataset. The mode holds in the case of Boolean features (e.g., symptom markers).

$$\hat{x} = \arg \max_{c \in \{0,1\}} \left(\sum_{i=1}^n \mathbf{1}\{x_i = c\} \right) \quad (10)$$

For numerical features (e.g., WBC, Platelet Count) the median is best for skewed distributions:

$$\tilde{x} = \text{median}(\{x_i \mid x_i \neq \text{NA}\}) \quad (11)$$

After handling missing values, the merged dataset proceeded on the same path of processing as the primary dataset: it went into the feature segregation step, went into the preprocessing pipeline, and then was applied to the custom Random Forest classifier. This single process guaranteed that both primary and merged datasets underwent the same steps of transformations and allowed both methodological consistency and the ability to compare their performance in a fair manner.

3.9 Performance Comparison

The performance of the models under consideration, specifically baseline pre-trained models was evaluated. The proposed custom Random Forest classifier was then compared against them in terms of model performance. All the models were trained and tested on the primary data with identical conditions and metrics considered, such as accuracy, precision, recall, F1-score, and AUC. The algorithm levels of Random Forest always surpassed baseline models, specifically in the case of recall and F1-score for the minority group.

The evaluation was conducted again with the merged dataset, applying the same procedure. The study demonstrated one more time that the custom Random Forest classifier model is superior and can be adapted to various datasets of various sources and distributions.

4 Result and Analysis

The result analysis compares the proposed model performance with the reference datasets. The results indicate its excellent predictive power and confirm the validity and usefulness of the suggested model.

4.1 Exploratory Data Analysis

An initial data analysis was done to obtain an overall idea about the dataset, with emphasis on the distribution of the categorical variables and its relation to the dengue diagnosis outcome. It was a necessary step, which would help recognize the trends and feature significance to perform further machine learning model implementation. In Fig. 3 the platelet count distribution is shown with a

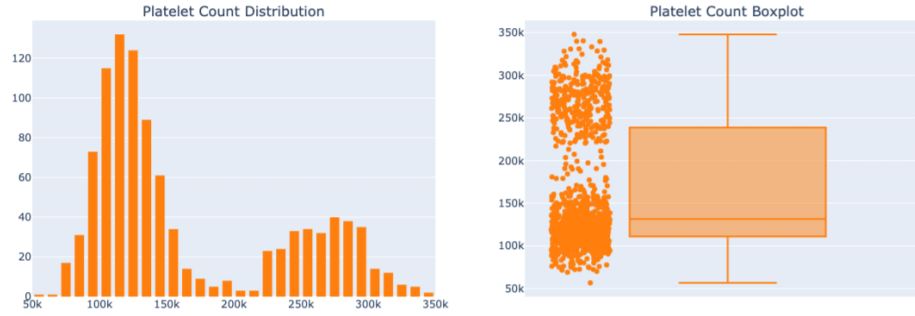


Fig. 3. Platelet Count histogram and boxplot overview.

great part of the patients having a concentration at between 100,000 and 150,000. The boxplot indicates that dengue-positive patients tend to have a lower platelet level, with some of them being lower than the normal healthy level.

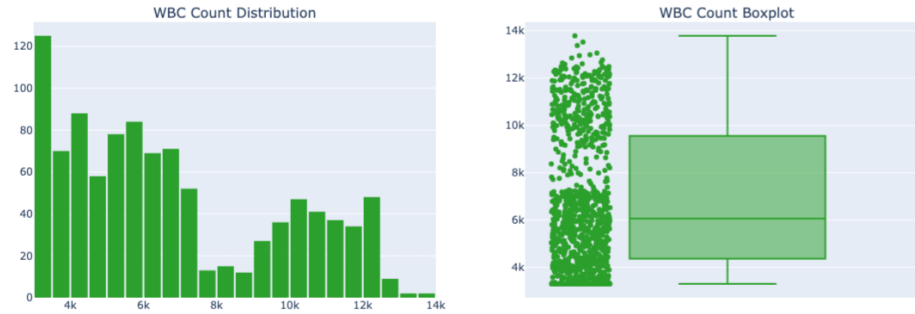


Fig. 4. WBC count histogram and boxplot overview.

The distribution of the WBC counts displayed in Fig. 4 indicates that a big percentage of patients could be counted in the range of between 3,000 and 7,000. The boxplot shows that the dengue-positive patients tend to have lower counts of WBC than those present in healthy individuals, which is clinically expected of dengue. Fig. 5 identifies the relationship between the presence of fever and

the outcome of dengue disease. Of all dengue positive, 659 individuals reported fever; in comparison, 38 had not. In negative tests, there was fever in 195 cases and no fever in 126 cases. That supports the high reliance of fever on dengue positivity, but not independently.

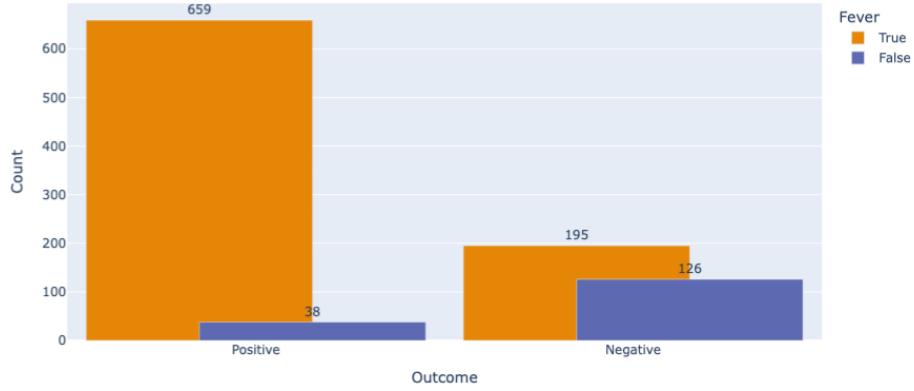


Fig. 5. Distribution of Positive and Negative Outcomes by Fever status

Analysis showed that low platelet count and WBC reduction were the best predictors of dengue, which is in line with clinical manifestations of the disease. There was also a strong correlation between fever and duration of fever, and this is characteristic of dengue. Other symptoms including headache, muscle pain, rash and vomiting were supportive predictive variables. In general, the main characteristics of the model are consistent with the commonly known clinical signs of dengue.

4.2 Model Performance

In this paper, dengue diagnosis has been benchmarked with six classifiers: LR, DT, RF, KNN, SVM and GBM. The last suggested model is an optimized Random Forest on the feature selection, preprocessing normalization, parameter tuning and Grid Search optimization, using stratified 5-fold CV. The models were all trained under the same conditions and were tested using precision, accuracy, recall, F1, and AUC on the primary dataset as well as on a harmonized merged dataset.

ROC curves in the Fig. 6 give a comparative analysis of the different machine learning classifiers on dengue diagnosis. In the left plot, the performance of six base models and a custom Random Forest model is shown using the primary dataset. Among them, custom Random Forest showed the best results with an AUC of 0.99, which reflects very strong discriminating power. Random Forest (AUC = 0.837) and Logistic Regression (AUC = 0.855) also did well whereas Decision Tree (AUC = 0.66), KNN (AUC = 0.73) and SVM (AUC = 0.73)

lagged behind with a less accurate predictive ability. The right plot demon-

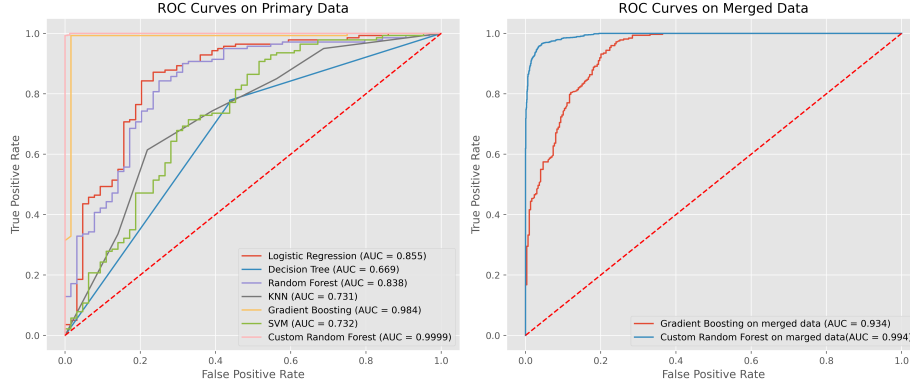


Fig. 6. ROC curve Comparison of Baseline and Custom Models Across Datasets.

strates the ROC curves of the Custom Random Forest and Gradient Boosting on a merged dataset containing both the primary and an external dataset. In Fig. 6 Custom Random Forest upheld an extremely high AUC of 0.99 which indicates an excellent generalization validity in various heterogeneous data, and Gradient Boosting retained its performance registering an AUC of 0.93. These findings demonstrate that, although various models yield high performance in working with the raw data, the Custom Random Forest model portrays the most stable and reliable predictive performance in combining various datasets, thus presenting a solid alternative when accessed to detect early dengue progression.

Table 2. Baseline & Comparative Model Performance (Original Dataset).

Model	Accuracy	Precision	Recall	F1-score	AUC
LR	82.35	0.87	0.88	0.87	0.855
DT	70.58	0.79	0.77	0.78	0.669
RF	82.84	0.85	0.91	0.88	0.838
KNN	70.09	0.81	0.74	0.77	0.731
SVM	77.45	0.79	0.91	0.85	0.732
GBM (on primary data)	98.43	0.99	0.99	0.99	0.984

Although GBM achieved perfect results on the primary dataset, its performance dropped to 84.95% accuracy with an AUC of 0.933 on the merged data, as shown in Table 3. In comparison, the suggested Custom RF achieved 90.49% accuracy and an AUC of 0.994, demonstrating better generalization. The custom Random Forest model was able to perfectly classify the original dataset and yet performed an excellent task of generalizing on the merged dataset. Model perfor-

Table 3. GBM on Merged data.

Model	Accuracy	Precision	Recall	F1-score	AUC
GBM (on merged data)	84.95	0.83	0.92	0.87	0.933

mance on the primary dataset was 99.02% accurate and the AUC was 0.99; thus, there was perfect discrimination. On the merged dataset the model performed at 90.49% accuracy on 2877 of 3,106 instances, with an AUC of 0.994 (Fig. 6), ex-tremely good generalization to diverse data.

Table 4. Proposed Custom Random Forest Model Performance.

Model	Accuracy	Precision	Recall	F1-score	AUC
Custom RF (on primary Dataset)	99.02	0.99	0.99	0.99	0.999
Custom RF (on merged Dataset)	90.49	0.88	0.99	0.93	0.994

5 Conclusion and Future work

The study presented a novel optimized Random Forest classifier including feature selection, class balancing with SMOTE, standardized preprocessing, hyper-parameter tuning to the Grid Search and the 5-fold stratified CV. The model was better in predictive performance than any of the baseline classifiers. On the primary data, it scored 99.02 accuracy and an AUC of 0.99, indicating perfect discrimination. Tested on merged and harmonized dataset, the model had a high generalization capacity of 90.49% accuracy and an AUC of 0.994.

Limitation. This evaluation is limited by its country-specific dataset for Bangladesh and thus findings may not generalize fully for regions with other populations, diseases, or infrastructures. Verification by multi-country validation is needed for broader use.

Future work. The model further extends to the prediction of a disease severity (e.g., DHF, DSS) rather than binary classification. Further research is needed to understand how to create multimodal (XGBoost, deep neural networks, temporal models such as LSTM) methodologies that would integrate clinical, genetic, and environmental features and make them more resilient and applicable across different populations. It is possible to adopt the model in the form of a mobile or web-based application, where healthcare professionals input basic symptoms and lab results (platelet, WBC) to generate a dengue risk score. This can be used to quickly and cheaply screen in low resource settings to help prioritize high-risk patients to confirmatory testing and follow-up.

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